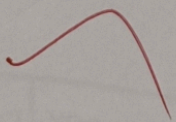
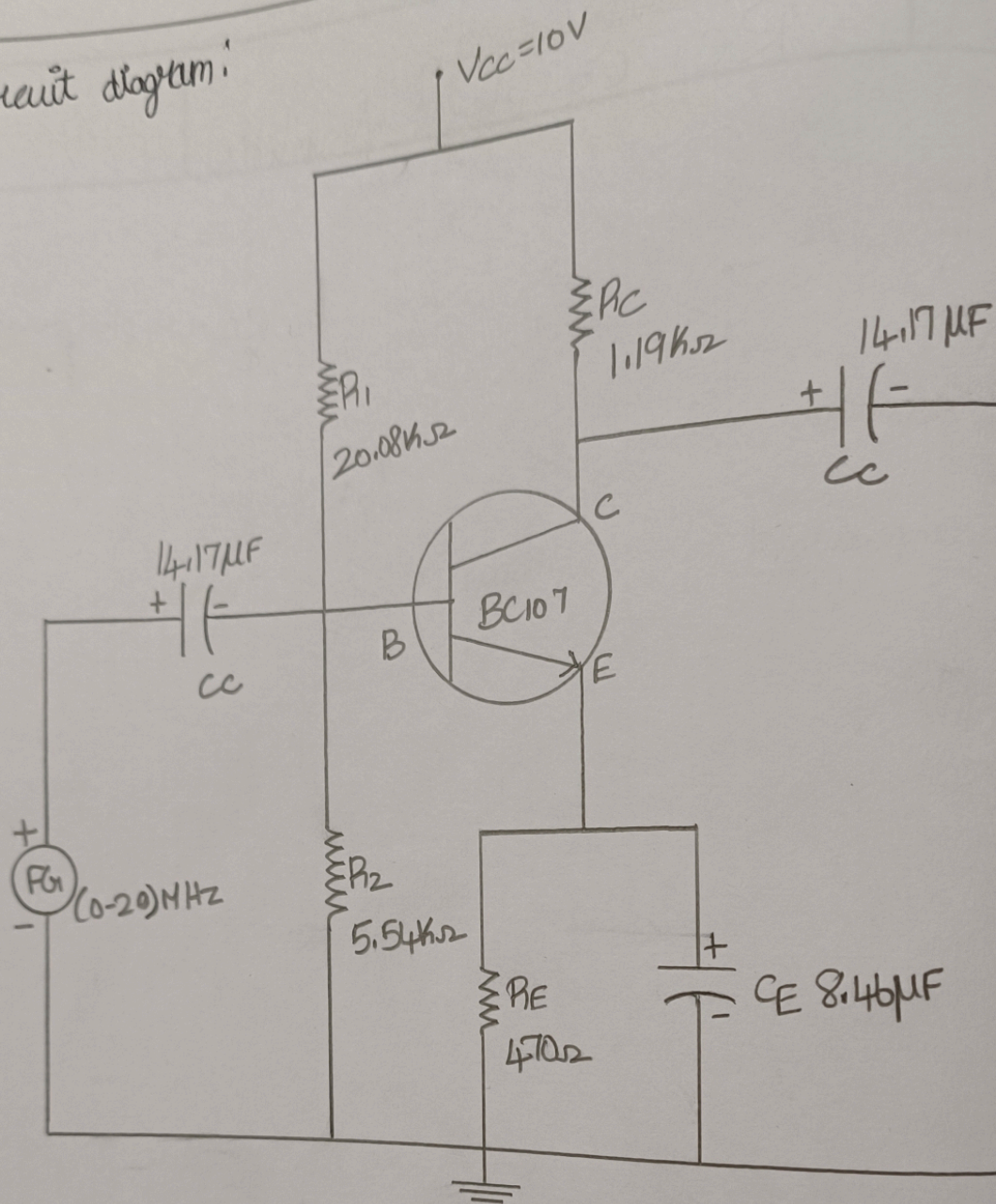


Circuit diagram:



ET 10/10/2020

| To design and m...     |                          |
|------------------------|--------------------------|
| Building capacity -    |                          |
| Components / Apparatus |                          |
| S.NO                   | Components               |
| 01                     | Resistor                 |
| 02                     | Capacitor                |
| 03                     | Transistor               |
| 04                     | Regulated power supply   |
| 05                     | Signal generator         |
| 06                     | Common emitter amplifier |
| 07                     | Capacitor                |
| 08                     | Resistor                 |
| 09                     | Transistor               |
| 10                     | Regulated power supply   |



Ex No: 03

Common emitter amplifier

RC Coupled Amplifier

11/05/24

Aim:

To design and measurement of frequency response, signal handling capacity, input and output impedances of RC coupled amp.

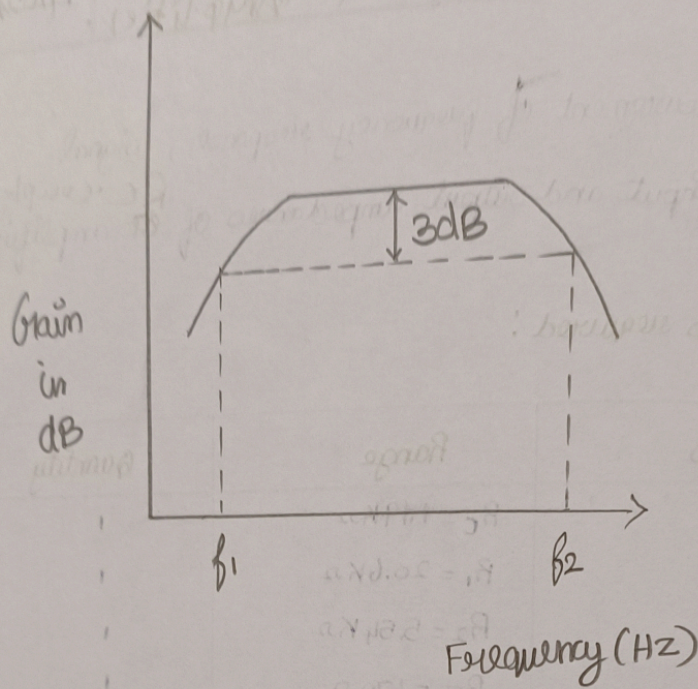
Components / Apparatus required:

| S.No | Components             | Range                | Quantity    |
|------|------------------------|----------------------|-------------|
| 01   | Resistor               | $R_C = 1.19 K\Omega$ | 1           |
|      |                        | $R_1 = 20.6 K\Omega$ | 1           |
|      |                        | $R_2 = 5.54 K\Omega$ | 1           |
|      |                        | $R_E = 470\Omega$    | 1           |
|      |                        | $R_L = 11.9 K\Omega$ | 1           |
| 02   | Capacitor              | $C_C = 14.17 \mu F$  | 2           |
|      |                        | $C_E = 8.46 \mu F$   | 1           |
| 03   | Transistor             | BC107                | 1           |
| 04   | Regulated power supply | (0-30)V              | 1           |
| 05   | Bread Board            | -                    | 1           |
| 06   | Connecting wires       | -                    | As required |
| 07   | CRO                    | (0-20) MHz           | 1           |
| 08   | Function Generator     | (0-2) MHz            | 1           |
| 09   | BNC probe              | -                    | 2           |

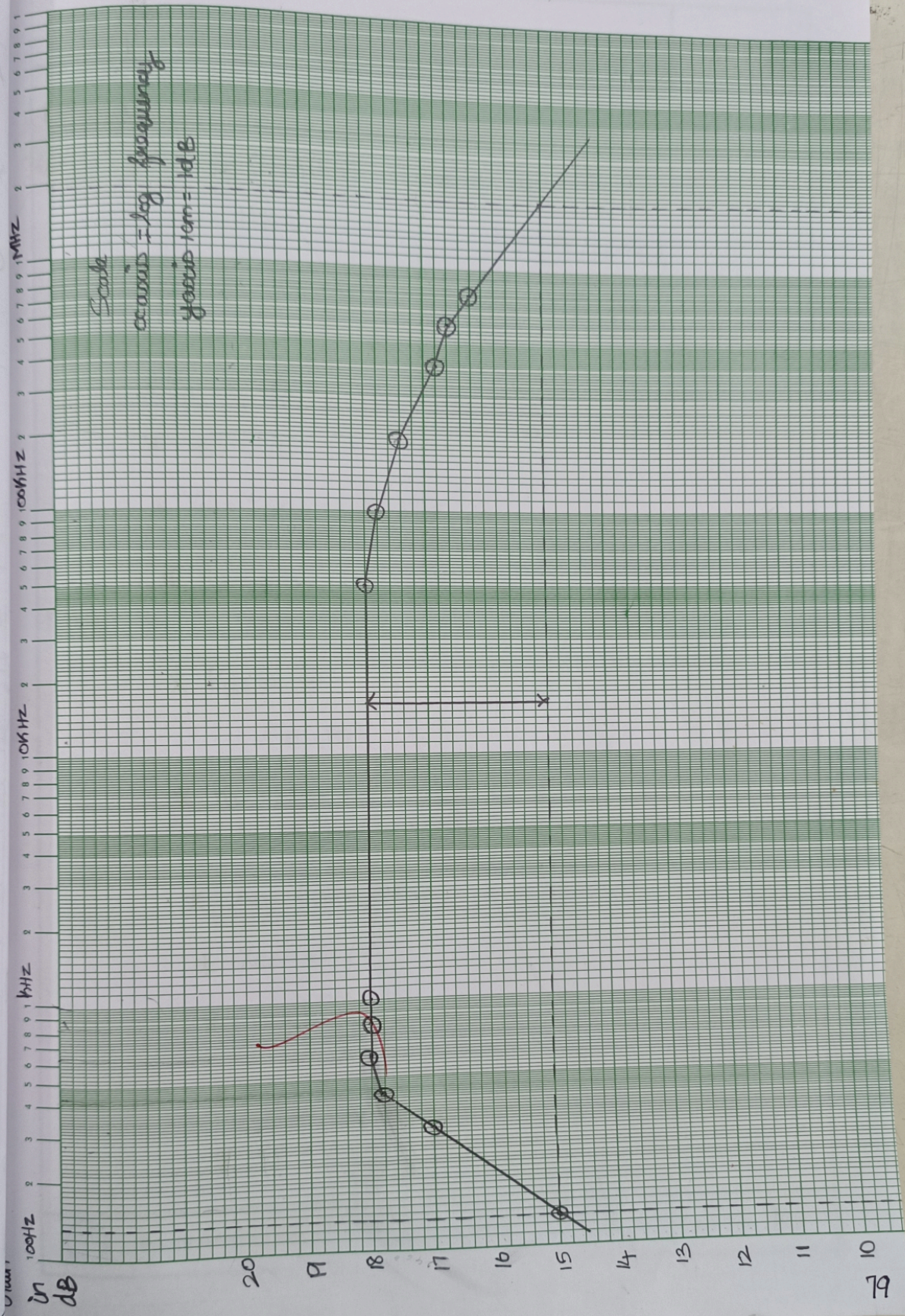




# Model Graph





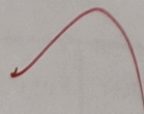




Tabulation :

$$V_{in} = 1 \times 20 \text{ mV}$$

| Frequency (Hz) | $V_{out} (V)$              | Gain = $\frac{V_{out}}{V_{in}}$ | Gain in dB<br>$20 \log (V_{out}/V_{in})$ |
|----------------|----------------------------|---------------------------------|------------------------------------------|
| 300 Hz         | $7.6 \times 20 \text{ mV}$ | 7.6                             | 17.616                                   |
| 400 Hz         | $7.8 \times 20 \text{ mV}$ | 7.8                             | 17.841                                   |
| 600 Hz         | $8 \times 20 \text{ mV}$   | 8                               | 18.061                                   |
| 800 Hz         | $8 \times 20 \text{ mV}$   | 8                               | 18.061                                   |
| 1 kHz          | $8 \times 20 \text{ mV}$   | 8                               | 18.061                                   |
| 50 kHz         | $8 \times 20 \text{ mV}$   | 8                               | 18.061                                   |
| 100 kHz        | $7.8 \times 20 \text{ mV}$ | 7.8                             | 17.841                                   |
| 200 kHz        | $7.4 \times 20 \text{ mV}$ | 7.4                             | 17.384                                   |
| 400 kHz        | $7.0 \times 20 \text{ mV}$ | 7.0                             | 16.901                                   |
| 600 kHz        | $6.8 \times 20 \text{ mV}$ | 6.8                             | 16.650                                   |
| 800 kHz        | $6.4 \times 20 \text{ mV}$ | 6.4                             | 16.123                                   |
| 1 MHz          | $6.2 \times 20 \text{ mV}$ | 6.2                             | 15.847                                   |



\*The single stage bipolar junction amplifier, the base terminal, the emitter terminal and the collector terminal. The basic circuit shown below.

\*The single stage bipolar junction amplifier.



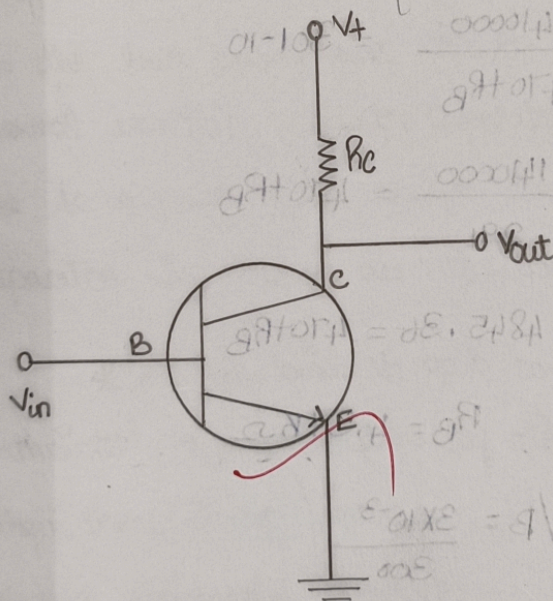
Theory:

## RC Coupled Amplifier.

Common emitter amplifier:

\*The common emitter amplifier is a basic single stage bipolar junction transistor and is used as a voltage amplifier. The input of this amplifier is taken from the base terminal, the output is collected from the collector terminal and the emitter terminal is common for both the terminal. The basic symbol of common emitter amplifier is shown below.

\*The common emitter amplifier is a basic single stage bipolar junction transistor and is used as a voltage amplifier.





## RC Coupled Amplifier

Working of common emitter amplifier:-

\*The below circuit diagram shows the working of the common emitter amplifier circuit and it consists of voltage divider biasing used to supply the basic bias voltage as per the necessity. The voltage divider biasing has a potential divider with two resistors are connected in a way that the midpoint is used for supplying base bias voltage.

\*There are differential types of electronic components in the common emitter amplifier which are  $R_1$  resistor is used for forward bias, the  $R_2$  resistor is used for the development of bias. The  $R_L$  resistor is used for output it is called as the load resistance. The  $R_F$  resistor is used for the thermal stability. The  $C_c$  capacitor is used to separate the AC signals from the DC biasing voltage and the capacitor is known as the coupling capacitor.

\*That the bias  $V_B$  gain common emitter amplifier transistor characteristics if the  $R_2$  resistor increase then there is an increase in the forward bias and  $R_1$  bias are universally proportional to each other.





\* In the common emitter amplifier and output resistors are medium.

Procedure : ✓

\* Connections are given as per the circuit diagram.

\* Set  $V_{in} = 20\text{mV}$  in the function generator keeping input voltage constant vary the frequency from  $100\text{Hz}$  to  $1\text{MHz}$  in regular steps.

\* Note down the corresponding output voltage.

\* Plot the graph, gain in dB  $\times$  frequency in Hz.

\* Calculate the Bandwidth from the frequency response graph.





Result:

Amplifier

R.C. Coupled Amplifier

Thus the ~~common emitter~~ amplifier was obtained and the frequency response was determined.

